

Business assumptions

Average purchase value (\$)  
20.00

Cost of sending the better coupon (\$)  
2.00

Targeting rule  
☒ Value threshold  
☐ Budget cap

Filter held-out customers

Coupon type

Bar

Carry out & Take a...

Coffee House

Restaurant(20-50)

Restaurant(<20)

Destination

Home

No Urgent Place

Work

Income bracket

\$100000 or More

\$12500 - \$24999

\$25000 - \$37499

\$37500 - \$49999

\$50000 - \$62499

\$62500 - \$74999

\$75000 - \$87499

\$87500 - \$99999

Less than \$12500

Causal ML for Coupon Targeting

Which customers should receive a 1-day coupon instead of a 2-hour coupon? Every reported causal estimate and targeting result below is computed on a held-out evaluation set, not the rows used to train the models.

Overview & Diagnostics   Top Customers   Customer Lookup

Showing 3,153 filtered customers from a 3,153-row held-out evaluation set.

| Customers to target | Incremental revenue | Campaign cost | Campaign ROI |
|---------------------|---------------------|---------------|--------------|
| 2,364               | \$8,485             | \$4,728       | 79.5%        |

Held-out causal estimates

| T-learner average uplift  | Propensity matching ATT   | Doubly-robust AIPW ATE    |
|---------------------------|---------------------------|---------------------------|
| +0.148                    | +0.171                    | +0.170                    |
| ↑ 95% CI [+0.129, +0.159] | ↑ 95% CI [+0.120, +0.229] | ↑ 95% CI [+0.132, +0.200] |

The estimators use different assumptions. Agreement is reassuring, but it does not by itself prove that all confounding has been removed.

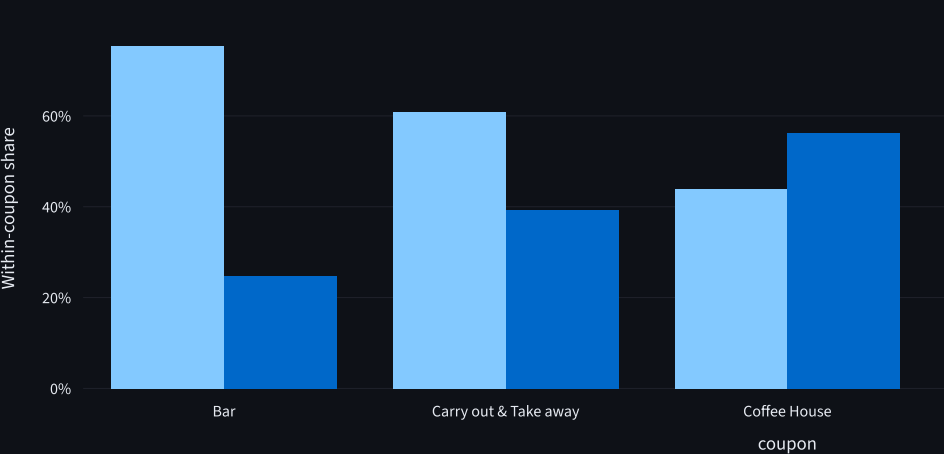
Positivity, overlap, and treatment-balance checks

| Control share (2h) | Treated share (1d) | Propensities clipped |
|--------------------|--------------------|----------------------|
| 44.2%              | 55.8%              | 1 (0.0%)             |

No coupon type exceeds 90% concentration in a single expiration arm.

Some held-out rows reached the 0.01/0.99 propensity clipping bounds and therefore have weak empirical counterfactual support.

Expiration mix within each coupon type



Propensity-score matching quality

| PSM match rate | Treated unmatched | Max  SMD  before | Max  SMD  after |
|----------------|-------------------|------------------|-----------------|
|----------------|-------------------|------------------|-----------------|

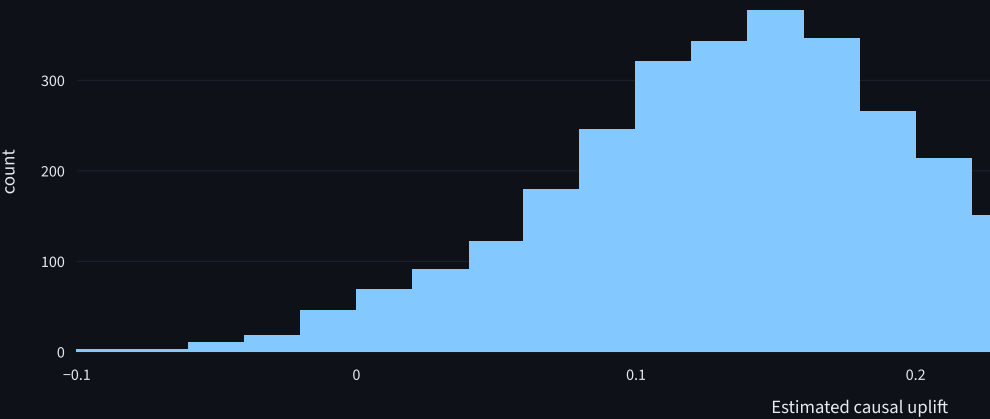
96.0%710.3310.237

Caliper: 0.20 SD of logit propensity score (width=0.185).

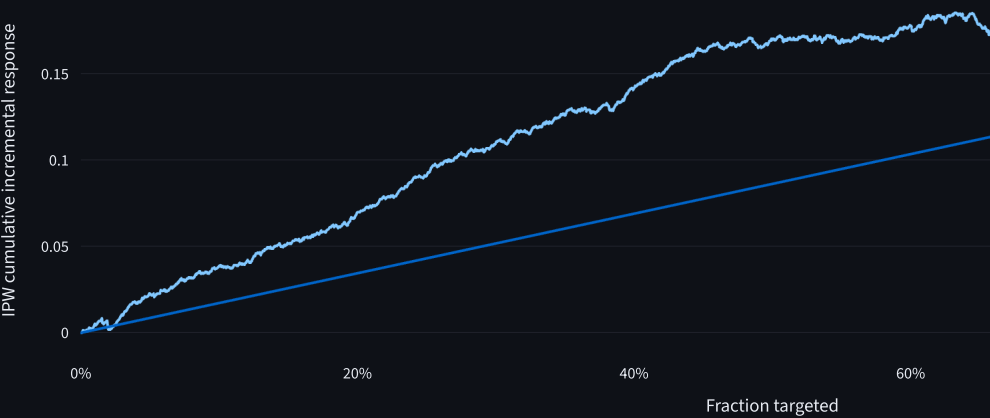
11 features still have |SMD| > 0.10 after matching.

|    | feature                  | smd_before | smd_after |
|----|--------------------------|------------|-----------|
| 18 | time_10PM                | -0.126     |           |
| 15 | passanger_Partner        | -0.05      |           |
| 3  | direction_same           | -0.098     |           |
| 19 | time_2PM                 | 0.226      |           |
| 12 | destination_Work         | 0.105      |           |
| 21 | time_7AM                 | 0.105      |           |
| 28 | maritalStatus_Single     | 0.065      |           |
| 16 | weather_Snowy            | 0.194      |           |
| 24 | coupon_Restaurant(20-50) | 0.113      |           |
| 25 | coupon_Restaurant(<20)   | -0.129     |           |

Held-out distribution of estimated individual uplift



Held-out uplift/Qini-style cumulative gain



Held-out average uplift by coupon type

